

Ava Schrandt, Software Engineer | Backend • Data Platforms • AI Systems

Frederick, Maryland, 765-543-5002, avaschrandt@gmail.com

EXPERIENCE

May 2025 — Present	Software Engineer — ClearSky Smart Fleet, Connected Solutions, JLG INDUSTRIES	Hagerstown, MD
	- Engineer internal Python/PyQt tooling that automates end-to-end validation of the CS550 ClearSky Smart Fleet device, tracing behavior from embedded firmware and CAN communication through backend services to cloud-reported telemetry. - Build CAN-based automated test and debugging workflows using PCAN, correlating live device traffic with cloud data to diagnose issues across embedded, backend, and connected-system boundaries. - Integrate testing infrastructure with Java backend microservices and package the tooling as a Windows executable for use across internal engineering and validation workflows. - Design and maintain time-series telemetry data pipelines using Azure Data Explorer (ADX), developing new feature data flows and troubleshooting production data issues supporting customer-facing ClearSky Smart Fleet applications. - Develop an internal AI assistant over enterprise data using Microsoft AI Foundry, implementing agent orchestration, retrieval, and validation with Python/FastAPI, TypeScript/React, and vector databases.	
May 2024 — May 2025	Software Engineer Intern — ClearSky Smart Fleet, JLG INDUSTRIES	
	- Initiated development of an automated end-to-end testing framework for the CS550 ClearSky Smart Fleet device, building Python/PyQt tooling to parse live CAN data and support device-to-cloud validation. - Enhanced existing internal engineering tools with secure Azure authentication and automated application setup and deployment workflows.	

TECHNICAL SKILLS

Python | Java | C++ | C | SQL | TypeScript | JavaScript | FastAPI | REST APIs | Java microservices | Kubernetes | Microsoft Azure | Azure Data Explorer (ADX) | NoSQL | MongoDB | Microsoft AI Foundry | Agent Orchestration | Automated end-to-end testing | PyQt | CAN bus | Windows application packaging | Git | Agile/Scrum | Jira | Confluence | Linux/Unix | Vector Databases | Jama Software

EDUCATION

Present	B.S. Computer Science — Software Engineering PURDUE UNIVERSITY	West Lafayette, IN
	GPA 3.62 • Dean's List: Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025	

SELECTED PROJECTS

May 2025	SafeEats	Built an iOS application for finding nearby dining options based on dietary restrictions using Python, FastAPI, MongoDB, Google Places, and Swift.
May 2024	Unix Shell Interpreter	Implemented shell behavior and command processing in C/C++ using Lex and Yacc.
May 2024	Serenova Sleep Tracker	Scrum Master for an Agile iOS team; developed features with SwiftUI, Firebase, and Apple HealthKit.